

STRIX V2 / TorqueEFI — STM32F411 Black Pill Pin Allocations

WeAct Studio Black Pill · STM32F411CEU6 · USB CDC tuner · 4-cyl sequential capable · Document generated for wiring reference

Outputs (actuators)

Function	Pin	Resource	Notes
Ignition 1	PB0	TIM3_CH3 / GPIO	Logic IGBT / smart coil
Ignition 2	PB1	TIM3_CH4 / GPIO	Logic IGBT / smart coil
Ignition 3	PB2	GPIO	Logic IGBT / smart coil
Ignition 4	PB3	GPIO	Logic IGBT / smart coil
Injector 1	PB15	TIM1_CH3N / GPIO	Low-side MOSFET (was PB4)
Injector 2	PB5	TIM3_CH2 / GPIO	Low-side MOSFET
Injector 3	PB6	TIM4_CH1 / GPIO	Low-side MOSFET
Injector 4	PB7	TIM4_CH2 / GPIO	Low-side MOSFET
Boost solenoid	PB8	TIM4_CH3 / PWM	Low-side MOSFET
Fan relay	PB9	GPIO	Flyback protected
Fuel pump relay	PB10	GPIO	Flyback protected
ETB PWM (H-bridge)	PA8	TIM1_CH1	Electronic throttle / idle
ETB DIR	PA9	GPIO	H-bridge direction
VVT1 intake	PA10	TIM1_CH3 PWM	Cam solenoid
VVT2 exhaust	PB14	TIM1_CH2N PWM	Cam solenoid

Inputs (sensors & triggers)

Function	Pin	Resource	Notes
Crank position	PA0	TIM5_CH1 IC	VR/Hall conditioned 0–3.3 V
MAP	PA1	ADC1_IN1	0–3.3 V scaled
TPS	PA2	ADC1_IN2	0–3.3 V scaled
CLT / ECT	PA3	ADC1_IN3	NTC + pull-up
IAT	PA4	ADC1_IN4	NTC + pull-up
O2	PA5	ADC1_IN5	NB or WB 0–3.3 V
Knock / spare	PA6	ADC1_IN6	Optional
Battery sense	PA7	ADC1_IN7	Divider → 3.3 V max
Cam 1	PA15	TIM2_CH1 IC	720° phase
Cam 2	PB4	TIM3_CH1 IC	Second cam (freed from INJ1)
Pedal position	PB12	ADC1_IN12	DBW pedal
Clutch switch	PB13	GPIO / EXTI	Launch / FFS

STRIX V2 — System & notes

System / debug / power

Function	Pin	Resource	Notes
USB D− / D+	PA11 / PA12	USB FS	CDC virtual COM (tuner)
SWDIO / SWCLK	PA13 / PA14	SWD	ST-Link programming
User LED	PC13	GPIO	Status (active low typical)
HSE 25 MHz	PH0 / PH1	OSC	Main crystal
3V3 / GND / 5V	—	Power	Do not feed 5 V into ADC/logic pins

Critical allocation notes

- **INJ1 = PB15** (not PB4). PB4 is reserved for **Cam 2** (TIM3_CH1).
- Crank on **PA0 / TIM5**, Cam1 on **PA15 / TIM2** — do not reassign without firmware changes.
- USB PA11/PA12 is the only tuner link (CDC). Keep ST-Link on PA13/PA14 for programming.
- All ADC inputs must be scaled to **0–3.3 V**. Battery via resistor divider.
- Injector/ignition drivers need external MOSFETs / IGBT or smart-coil drivers; MCU is logic-level only.
- Relays (fan, FP) need flyback diodes. Solenoids (boost, VVT) need freewheel protection.
- CubeMX project: **STRIX_V2.ioc** · Linker: **STM32F411CEUX_FLASH.ld** (384 KB app + NVM sector 7).
- Source of truth: **Core/Inc/ecu_pins.h** in the firmware tree.

Quick group index

Group	Pins
Crank / cams	PA0, PA15, PB4
Analog sensors	PA1–PA7, PB12
Ignition 1–4	PB0–PB3
Injection 1–4	PB15, PB5, PB6, PB7
PWM / VVT / ETB	PA8, PA9, PA10, PB14, PB8
Relays	PB9, PB10
USB / SWD	PA11–PA14

STRIX V2 · TorqueEFI · For laboratory / motorsport development only · Verify against your PCB revision before wiring